

Software Requirement Specification

1. Introduction

1.1 Overview

This Software Requirement Specification (SRS) describes the requirements for the Digital Examination Attendance System (Sommerville, 2016). The purpose for this system is to improve examination attendance verification, track examination script submission, confirm student's identity and manage records for the students in the University of Eswatini. This document will benefit the system developers, lecturers, invigilators, testers and maintenance personnel. This SRS document provides a detailed description of the functional and non-functional requirements of the based on (Engineers, 1998).

1.2 Scope

The Digital Examination Attendance Verification System is a web-based system designed to substitute the traditional paper-based examination cards. This system is expected to do the following functions:

- Electronically record the examination attendance
- Approve the identity of a student digitally
- Trace the examination script submission
- Securely store the attendance records
- Produce an examination attendance report

The system will not do the following:

- Mark the examination scripts
- Replace the registration system of the University
- Grade the students

Objectives of the system:

- Reduce the loss and damage of examination attendance records

- Improve student identity verification
- Provide proof of examination script submission
- Eliminate manual counting of examination cards
- Improve accountability and traceability in the examination processes

1.3 Definitions, Acronyms and Abbreviations

Term	Meaning
SRS	Software Requirement Specification
DFD	Data Flow Diagram
UML	Unified Modeling Language
Admin	System administrator
Invigilator	Examination supervisor
Verification	Process of confirming student identity
Authentication	Process of validating user credentials
Database	Organized collection of data
Attendance Record Digital	Proof that a student attended an examination

1.4 References

References

- Dube, E. L. (2022). *Tools and Techniques Review Lecture Note 01*. Matsapha: Enock Lukhetfo Dube.
- Engineers, T. I. (1998). *IEEE Recommended Practice for Software Requirements Specifications*. New York.
- Sommerville, I. (2016). *software Engineering*. Harlow, England: Pearson.

1.41 Overview

This document is organized as follows:

- Section 1 Introduces the system
- Section 2 Describes the overall system
- Section 3 Explains the specific functional and non-functional requirements
- Section 4 contains appendices and supporting information

2 Overall Description

2.1 Product Perspective

The digital Examination Attendance Verification System is a new independent software product that will support examination management processes in the University of Eswatini.

Currently the examination process relies on physical examination cards, which creates several problems including:

- loss or damage of examination cards
- weak identity verification
- lack of proof of script submission
- human errors during counting of attendance
- poor accountability and traceability

The proposed system introduces a centralized digital platform where invigilators can verify student identity, record attendance electronically, confirm examination script submission and generate reports automatically. The system will interact with the student database, examination records database and user authentication system. According to (Engineers, 1998) the product perspective explains how the software relates to larger systems and interfaces.

2.2 Product Functions

The major functions (function view) of the system are:

- User authentication and login
- Student identity verification
- Digital attendance recording
- Examination script submission confirmation
- Attendance report generation
- Examination record storage
- Search and retrieval of attendance records
- User management
- Audit trial generation

2.3 User Characteristics

Administrator- Manages system users, maintains system records, generates reports, and requires moderate computer skills

Invigilator- verifies student attendance, confirms script submission, and requires basic computer skills

Lecturer- views attendance reports and monitors examination records

Student- presents identification during verification and views attendance confirmation

2.4 Constraints

The system shall operate with the constraints of internet connectivity being available during examinations, users must have authorized accounts, the system should comply with the university examination policies, the system is expected to protect confidential student data and the system must operate on the university's available hardware. This is according to the (Engineers, 1998) that constraints can include policies, interfaces, hardware limitations and reliability requirements.

2.5 Assumptions and Dependencies

Assumptions- students are registered before examination begins, invigilators are trained to use the system and examination venues have access to computers or mobile devices.

Dependencies- availability of institutional student databases, availability of the internet and availability of server infrastructure.

3 Specific Requirements

3.1 Functional Requirements

This section describes the main functions that the Digital Examination Attendance Verification System shall provide.

FR1: User Authentication

The system shall allow authorised users to log in using valid credentials.

- The system shall require users to enter a username and password before accessing the system.
- The system shall deny access to users with invalid credentials.
- The system shall restrict system functions based on the type of user.
- The system shall allow authorised users to log out after using the system.

FR2: User Role Management

The system shall support different user roles.

The main user roles shall include:

- Administrator
- Invigilator
- Lecturer / Examiner

The administrator shall have full access to manage system data. The invigilator shall be allowed to scan student cards, confirm attendance, scan scripts, and view exam dashboards. The lecturer or examiner shall be allowed to search for courses, view dashboards, and access reports.

FR3: Student Record Management

The system shall allow administrators to manage student records.

The system shall allow the administrator to:

- Add new student records.
- Update existing student details.
- Delete student records where necessary.
- Search for students using student number, card ID, or student name.
- Store student information such as student number, full name, card ID, programme, and photo.

FR4: Department and Course Management

The system shall allow administrators to manage departments and courses.

The system shall allow the administrator to:

- Add departments.
- Add courses under specific departments.
- Update department and course details.
- View courses grouped according to their departments.
- Search for courses using course code or course name.

FR5: Exam Session Management

The system shall allow administrators to create and manage exam sessions.

Each exam session shall include:

- Course
- Exam date
- Start time
- End time
- Venue

The system shall use exam session information to determine which paper is active when a student scans at the door.

FR6: Student Exam Registration

The system shall allow administrators to register students for specific exam sessions.

The system shall:

- Link a student to one or more exam sessions.
- Prevent duplicate registration of the same student for the same exam session.
- Use registration records to check whether a student is allowed to write a particular exam.
- Deny access if a student is not registered for the detected active exam session.

FR7: Central Student Entry Scanning

The system shall provide a central student entry scanner.

The system shall allow an invigilator to scan or enter a student card ID at the exam venue entrance.

After scanning, the system shall:

- Identify the student using the scanned card ID.
- Check the current date and time.
- Detect the active exam session based on the timetable.
- Verify that the student is registered for the detected exam session.
- Display the student's details for confirmation.
- Display the detected course, venue, time, and assigned seat number.
- Allow the invigilator to confirm the student's identity.
- Record the student as present after confirmation.

FR8: Student Identity Verification

The system shall assist invigilators in verifying student identity.

The system shall display:

- Student full name
- Student number
- Programme
- Student photo where available
- Course being written
- Seat number

The invigilator shall confirm that the student standing at the door matches the student record before the system records attendance.

FR9: Attendance Recording

The system shall record examination attendance electronically.

The system shall:

- Create an attendance record when a registered student is confirmed at the door.
- Store the date and time of the scan.
- Store the user who confirmed the attendance.
- Prevent duplicate attendance records for the same student and exam session.
- Show a warning if a student attempts to scan more than once for the same exam.

FR10: Venue Exam Block Management

The system shall group exam sessions according to venue, date, start time, and end time. The system shall allow multiple papers to be written in the same venue at the same time. For example, if CSC322, MAT321, and PAD345 are written in Main Hall from 09:00 to 12:00, the system shall treat them as one venue exam block for seating purposes.

FR11: Seat Allocation

The system shall generate seat numbers for students before the exam.

The system shall:

- Generate seat numbers for all students writing in the same venue and time block.
- Group students by course when assigning seats.
- Assign seat numbers in sequence.
- Ensure that each student has only one seat number for a specific exam session.
- Ensure that seat numbers are unique within the same venue exam block.
- Allow seat numbers to be viewed before the exam.

Example:

- CSC322 students may be allocated seats 001 to 050.
- MAT321 students may be allocated seats 051 to 080.
- PAD345 students may be allocated seats 081 to 120.

FR12: Seating Plan Export

The system shall allow authorised users to export the venue seating plan.

The seating plan shall include:

- Seat number
- Course code
- Course name
- Student number
- Student name
- Programme
- Script code
- Venue
- Exam date
- Start time
- End time

The exported seating plan shall be downloadable in CSV format so that invigilators can print it and place personalised scripts on the correct desks before the examination begins.

FR13: Script Record Generation

The system shall generate personalised script records for registered students.

Each script record shall be linked to:

- Student
- Exam session
- Unique script code
- Script status

The system shall allow missing script records to be generated before or during the exam.

FR14: Central Script Submission Scanning

The system shall provide a central script submission scanner.

The system shall allow an invigilator to scan or enter a script code when a student submits an examination script.

After scanning, the system shall:

- Identify the script using the unique script code.
- Identify the student linked to the script.
- Identify the exam session linked to the script.
- Display the student's details.
- Display the course, venue, seat number, and script code.
- Check whether the student was recorded at the door.
- Allow the invigilator to confirm script submission.
- Mark the script as submitted after confirmation.

FR15: Script Submission Verification

The system shall verify whether a submitted script belongs to the correct student and exam session.

The system shall:

- Reject unknown script codes.
- Warn the user if the script was already submitted.
- Mark a script as submitted only after invigilator confirmation.
- Flag a script if it is submitted by a student who was not recorded at the door.
- Store the submission time and the user who received the script.

FR16: Exam Dashboard

The system shall provide a dashboard for each exam session.

The dashboard shall display:

- Total expected students
- Total present students
- Total missing students
- Total submitted scripts
- Students present but script not submitted
- Scripts submitted without door scan
- Flagged attempts
- Seat allocation information where available

The dashboard shall allow invigilators, lecturers, and examiners to monitor the status of the exam.

FR17: Search Functionality

The system shall provide search features on the landing page.

The system shall allow authorised users to:

- Search for an exam session using course code or course name.
- Search for a student using student number, card ID, or student name.
- View the student's registered exams.
- View the student's attendance and script submission status.
- Access related dashboards from the search results.

FR18: Flagged Attempt Recording

The system shall record suspicious or invalid attempts.

The system shall record a flagged attempt when:

- An unknown student card is scanned.
- A student is not registered for the active exam.
- A student attempts to scan more than once.
- An invalid script code is scanned.
- A script is scanned more than once.
- A script is submitted without a matching door scan.

Each flagged attempt shall store the scanned value, exam session, reason, type of flag, and time of attempt.

FR19: Audit Report Generation

The system shall generate an audit report for each exam session.

The audit report shall include:

- Student number
- Full name
- Programme
- Door attendance status
- Door scan time
- Script submission status
- Script code

- Script submission time
- Flag reason where applicable

The report shall be exportable in CSV format.

FR20: Department-Based Exam Browsing

The system shall allow users to browse exam sessions by department.

When a department is selected, the system shall display all exam sessions under that department, grouped by exam date. Each exam session shall provide access to its dashboard, scanners, script records, and seating functions.

FR21: Duplicate Prevention

The system shall prevent duplicate records.

The system shall:

- Prevent the same student from being marked present more than once for the same exam session.
- Prevent the same script from being submitted more than once.
- Prevent duplicate student registration for the same exam session.
- Prevent duplicate seat allocation for the same student and exam session.

FR22: Confirmation Messages and Alerts

The system shall provide clear messages to users.

The system shall display:

- Success messages when attendance or script submission is recorded.
- Error messages when a student or script is not found.
- Warning messages for duplicate scans.
- Warning messages for unregistered students.
- Warning messages for scripts submitted without door scans.

These messages shall help users understand what action has been completed or what problem has occurred.

FR23: Record Storage

The system shall store all important examination records in the database.

The system shall store:

- Student records
- Department records
- Course records
- Exam session records
- Exam registration records
- Venue exam blocks
- Seat allocations
- Attendance records
- Script records
- Flagged attempts
- User accounts

FR24: Report Download

The system shall allow authorised users to download reports.

The system shall support downloading:

- Exam audit reports
- Venue seating plans

The downloaded reports shall be in CSV format so that they can be opened using spreadsheet software such as Microsoft Excel or Google Sheets.

3.2 Non-Functional requirements

Performance requirements: The system shall respond to user requests within 5 seconds and the system shall support multiple users simultaneously during examinations.

Security Requirements: Only authorized users shall access the system, student information shall be protected from unauthorized access and passwords shall be encrypted.

Reliability Requirements: The system shall store records without loss and securely and backup mechanisms shall be available.

Usability Requirements: The system interface shall be simple and easy to use, and minimal training will be required on users.

Maintainability Requirements: The system will be easy to update and to maintain and errors should be easy to identify and correct.

Availability requirements: The system shall be available during examination period and this shall minimize downtime.

3.3 External Interface Requirements

User Interfaces

The system shall provide the following user interfaces:

- Login Interface: The system shall provide a secure login page where authorised users such as administrators, examiners, lecturers, and invigilators can access the system using their credentials.
- Examination Control Centre / Landing Page: The system shall provide a main landing page that acts as the examination control centre. This page shall allow users to access the central student entry scanner, central script submission scanner, today's exam sessions, department-based exam browsing, course search, and student search.

- Department and Course Browsing Interface: The system shall allow users to browse examination sessions by department and view all papers scheduled under a selected department. Exam papers shall be grouped by date to make navigation easier.
- Course Search Interface: The system shall provide a search feature that allows an examiner or lecturer to search for an examination paper using the course code or course name. The results shall show the related exam session, venue, date, time, and dashboard access.
- Student Search Interface: The system shall provide a search feature that allows an authorised user to search for a student using the student number, card ID, or name. The result shall show the student's registered exam sessions, attendance status, script submission status, and related dashboard links.
- Central Student Entry Scanner Interface: The system shall provide a central scanner page where invigilators can scan or enter a student card ID. The system shall automatically identify the student, determine the active exam session based on the timetable, verify that the student is registered for that exam, display the student's details and photo where available, and show the assigned seat number before attendance is confirmed.
- Central Script Submission Scanner Interface: The system shall provide a script scanner page where invigilators can scan or enter a personalised script code. The system shall automatically detect the student, the exam session, the script status, and the related seat number. It shall also verify whether the student was recorded at the door before accepting the script submission.
- Exam Dashboard Interface: The system shall provide a dashboard for each examination session. The dashboard shall show expected students, present students, missing students, submitted scripts, students present without submitted scripts, scripts submitted without door scans, and flagged attempts.

- Venue Seating Plan Interface: The system shall provide a seating plan page for each venue and time block. This page shall show all students writing different papers in the same venue at the same time, with seat numbers grouped by course.
- Report and Export Interface: The system shall allow authorised users to export audit reports and venue seating plans in CSV format. These reports shall support invigilators and examiners in verifying attendance, script collection, and seating arrangements.
- Administration Interface: The system shall provide an admin interface for managing departments, courses, students, exam sessions, student registrations, script records, seat allocations, attendance records, and flagged attempts.

The user interfaces shall be simple, clear, and suitable for use during examination periods. Since exam environments are time-sensitive, the system shall use clear buttons, status badges, warning messages, and confirmation prompts to reduce user mistakes.

(Engineers, 1998) recommends that user interfaces describe how users interact with the software.

Hardware Interfaces

The system shall support:

- Desktop computers
- Laptops
- Tablets
- QR code scanners
- Barcode scanners
- Student ID cards
- Printed personalised answer scripts with QR/barcode values
- Printers for printing seating plans and reports

The QR or barcode scanner shall function as an input device. When a student card or script code is scanned, the scanned value shall be entered into the appropriate scanner input field.

The system shall then process the value and retrieve the correct student or script record.

The system shall also support normal keyboard input, so that card IDs and script codes can be entered manually when a scanner is not available or when testing the prototype.

Software Interfaces

The system shall interface with:

- Django Web Framework: The system shall be implemented using Django, which shall handle routing, views, templates, authentication, form processing, database interaction, and server-side logic.
- Database Management System: The system shall use a relational database to store students, departments, courses, exam sessions, registrations, attendance records, scripts, seat allocations, venue exam blocks, and flagged attempts.
- Django Authentication System: The system shall use Django's authentication features to restrict access to authorised users only.
- Django Administration Panel: The system shall use the Django admin panel to allow administrators to manage system records such as students, courses, exam sessions, exam registrations, seat allocations, and scripts.
- Web Browser: The system shall be accessed through standard web browsers such as Google Chrome, Microsoft Edge, Mozilla Firefox, or any modern browser that supports HTML, CSS, and JavaScript.
- Bootstrap Framework: The system shall use Bootstrap for styling, layout, responsive design, buttons, cards, forms, tables, and alerts.

- CSV File Interface: The system shall generate CSV files for audit reports and venue seating plans. These files can be opened using spreadsheet software such as Microsoft Excel or Google Sheets.
- Operating System File System: The system shall allow exported reports and seating plans to be downloaded and stored on the user's computer.

Communication Interfaces

The system shall support:

- Local area network (LAN)
- Internet-based access

The system shall be accessible over a network using HTTP or HTTPS. In a local university environment, invigilators, lecturers, and administrators may access the system through a local area network. If deployed online, the system may also be accessed through the internet using a web browser.

The system shall support request and response communication between the client browser and the Django web server. User actions such as scanning cards, scanning scripts, searching courses, generating seating plans, and exporting reports shall be sent to the server for processing. The server shall then return the appropriate web page, confirmation message, warning message, report, or updated dashboard.

Where internet-based access is used, HTTPS is recommended to protect sensitive student and examination data during transmission.

4 Analysis and Modelling Techniques

The project will use the incremental modelling since our system is made up of separate modules that can be developed one at a time. This allows us to build and test each feature individually the authentication of the admin, attendance recording, and report

generation, before integrating them into the complete system. This approach reduced risk and improved system quality, especially because the university does not have an already existing software system to use as reference and to improve. According to (Dube, 2022), to develop this system we decided to use the standard method simple because this will save us time in developing new methods, and they are well documented. In the standard method we chose the Structure System Analysis and Structured design (SSA/SD), that included the Data Flow Diagram (DFD), Data Dictionary, and Structure English and the Object Modelling Techniques (OMT) which is a predecessor of the Unified Modelling Language, therefore we made use of the Use Case Diagram, Class Diagram, Sequence Diagram and the Activity diagram

5. System Models

5.1 Use Case Actors

- Invigilator
- Lecturer
- Student
- Administrator

5.2 Main Use Cases

- Administrator Login
- Record Attendance
- Confirm Script Submission
- Generate Reports
- Search Records

6. Future Enhancements

Possible future improvements include:

- Biometric verification
- Facial recognition
- Mobile application support
- Integration with student portals
- Real-time analytics dashboards

7. Conclusion

The Digital Examination Attendance Verification System will improve the efficiency, security, reliability, and accountability of the examination attendance process. The system addresses the weaknesses of the manual examination card method by introducing secure digital verification, attendance tracking, and script submission confirmation.

The proposed system aligns with software engineering principles discussed in IEEE SRS standards and modern software engineering methods described by (Sommerville, 2016).